

Project Management Report

Task Assignment, Risk Management & KPIs

Task Assignment & Roles

The following table outlines the distribution of roles and technical responsibilities among the team members to ensure a structured development workflow.

Name	Role	Technical Responsibilities
Abdelhalim	Team Leader	Project management, documentation writing, and tracking deliverables.
Youssef	Frontend Developer	UI design, API integration, and ensuring a seamless user experience (UX).
Abdulmajeed	Backend Developer	Building server-side logic, creating APIs using .NET 8, and data protection.
Mohamed	Database Engineer	Schema design, SQL Server management, and ensuring data query efficiency.

Risk Management Strategy

To ensure project continuity and technical excellence, the following risks have been identified along with their specific mitigation strategies.

Risk	Impact	Mitigation Strategy
API Integration Failures	High	• Adopting a Contract-first API design using Swagger. This allows frontend developers to work with Mock Data while backend logic is being finalized. • Conducting weekly Integration Testing sessions.
DB Performance Issues	Medium	• Schema design with optimized Indexing from day one. • Careful use of Entity Framework Core Tracking to prevent memory leaks, combined with regular Load Testing.
Timeline Slippage	High	• Following the MVP (Minimum Viable Product) methodology to prioritize core features. • Weekly Stand-up meetings to review the Gantt Chart and resolve bottlenecks immediately.
Data Security Breach	High	• Password hashing using BCrypt and securing all communications via HTTPS. • Implementing JWT (JSON Web Tokens) for secure session management.

Key Performance Indicators (KPIs)

These indicators represent the technical commitments and quality standards upheld by the development team.

1. API Response Time (Performance)

- **Target:** Average response time less than **500ms** per request.
- **Measurement:** Monitoring through tools such as **Postman** or **Application Insights**.

2. System Uptime (Availability)

- **Target:** **99%** system availability.
- **Measurement:** Utilizing stable cloud hosting and periodic audits of **IIS Logs**.

3. Guide Approval Efficiency

- **Target:** Processing new guide applications in less than **24 hours**.
- **Measurement:** Calculating elapsed time from document upload to the "Approved" status update.

4. Booking Completion Rate

- **Target:** Exceeding **85%** successful booking attempts.
- **Measurement:** Analysis of transactional reports extracted from the database.

5. Quality Assurance (Bug Rate)

- **Target:** **Zero Critical Bugs** at the time of delivery.
- **Measurement:** Continuous **QA Cycles** covering all functional requirements.